Asymmetric Process-Reward Adaptation for Mathematical Reasoning

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Abstract

Correct final answers can conceal faulty derivations, making outcome labels noisy supervision for process reward models (PRMs). We propose asymmetric offline adaptation: retain failure supervision and public step labels while omitting the positive outcome term. Four matched GRPO experiments test whether this change improves small-model mathematical reasoning.

1 Motivation and research question

Mathematical RL needs rewards that distinguish valid reasoning from lucky answers. Yet answer-correct solutions can contain intermediate errors (Zheng et al., 2025), so positive outcome supervision may teach a PRM to certify flawed derivations. We ask: **can withholding this supervision improve downstream policy accuracy?** We hypothesize that less false certification can outweigh lost useful supervision, yielding better rewards without new annotations.

2 Related work and contribution

PRM800K supplies step labels (Lightman et al., 2023); PRIME learns implicit rewards online (Cui et al., 2025); PURE addresses reward hacking through min-form credit assignment (Cheng et al., 2025). LCA learns PRMs with softmax-weighted pooling and already studies one-sided label noise (Jia et al., 2026). Our symmetric control adopts its pooling. The proposed contribution is an **asymmetric supervision rule with retained step labels**, evaluated through downstream GRPO. The D–C comparison isolates the removed loss term; C is LCA-inspired, not a full reproduction.

3 Proposed method

Offline adaptation. Generate four solutions per problem on 512 training problems using the initial policy. Retain complete, unambiguously verified trajectories. For native step-boundary correctness scores $p_{\phi,t}$, set $e_{\phi,t} = 1 - p_{\phi,t}$. Following LCA,

$$w_t = \frac{\exp(e_{\phi,t}/T)}{\sum_j \exp(e_{\phi,j}/T)}, \qquad E_\phi(\tau) = \sum_t \mathrm{sg}(w_t) e_{\phi,t}, \qquad T = 0.2. \tag{1}$$

Here sg stops gradients through weights. Define $L_{\text{fail}} = \mathbb{E}_{\tau^-}[-\log E_\phi(\tau^-)]$ and $L_{\text{success}} = \mathbb{E}_{\tau^+}[-\log(1 - E_\phi(\tau^+))]$, taking separate means over failures/successes. A shared set of 2,048 public positive/negative step labels supplies mean binary cross-entropy L_{step}:

$$L_c = L_{\text{fail}} + cL_{\text{success}} + L_{\text{step}}, \qquad c_C = 1, \quad \boxed{c_D = 0 \text{ (ours)}}. \tag{2}$$

Failure labels provide weak evidence of an error, without locating it. Real step labels discourage blanket rejection; downstream gains remain an empirical question.

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Policy training. Freeze each PRM throughout RL. For complete, parseable responses in B–D, compute

$$R_i = y_i + 0.2 \min_t p_{\phi,t}(\tau_i), \qquad \hat{A}_i = \frac{R_i - \bar{R}}{\sigma_R + 10^{-4}}. \tag{3}$$

Here $y_i \in \{0, 1\}$ is rule-verified answer correctness; $\bar{R}, \sigma_R$ are the group mean and sample standard deviation. A uses $R_i = y_i$; truncated or unparseable responses receive zero reward in all arms. GRPO applies the detached sequence advantage to completion tokens (Shao et al., 2024). **Correct policy rollouts remain in RL and retain their outcome reward.**

4 Experimental design and expected findings

Arm	Reward and PRM preparation	Question tested
A	Outcome only; no PRM	Reference GRPO performance
B	Outcome + original frozen PRM	B–A: value of an existing PRM
C	Outcome + PRM adapted with $c = 1$	C–B: value of full adaptation
D (ours)	Outcome + PRM adapted with $c = 0$	**D–C: value of the proposed change**

Models and implementation. Use Qwen2.5-Math-1.5B as policy and the H4 1.5B PRM (Hugging Face H4, 2025), with LoRA ranks 16 and 8. TRL 0.23.1/PEFT uses `loss_type="grpo"` and sums reward components before group normalization (Hugging Face, 2026). The outcome-only training and evaluation pipeline has been verified on one 80 GB A100; PRM-based comparisons remain to be run.

Data and contamination. The audited MATH split contains 6,993 training and 500 validation problems (Hendrycks et al., 2021). Use **MATH500** for primary evaluation and GSM8K test for transfer (Cobbe et al., 2021). Neither has exact normalized question overlap with the public PRM800K training inventory. All newly selected anchors will exclude validation, test, and diagnostic questions. Since PRM800K incorporates 4,500 original MATH test questions (OpenAI, 2023), full MATH is supplementary only. Unknown pretraining exposure and paraphrases remain limitations.

Matched controls and budget. C/D share PRM initialization, batches, optimizer, and update count; only c changes, without re-averaging the remaining losses. All arms share policy initialization, prompt order, decoding, and tuning effort: eight rollouts/question, temperature 1.0, and a 2,048-token cap. Run four arms with three paired seeds at **24M generated completion tokens/run** (288M total), retaining 6M/12M/24M milestones. Count incorrect and truncated completions; report update counts, prompt counts, response lengths, and end-to-end GPU hours, including PRM costs.

Evaluation and interpretation. Evaluate the initial policy and all three milestones with identical greedy decoding, without PRM reranking or voting. The primary comparison is **D–C MATH500 accuracy at 24M tokens**; D–A/B measures practical value. Report per-seed differences, mean/std, and 95% paired question-cluster bootstrap intervals, keeping all arm/seed predictions for a question together. Intermediate scores measure learning progress and do not select the final checkpoint.

Mechanism and risks. On fixed ProcessBench MATH solutions, measure first-error accuracy, correct-chain acceptance, and their harmonic mean at threshold 0.5. Within answer-correct solutions, valid-chain AUROC tests whether the PRM distinguishes valid from flawed reasoning beyond a uniform score shift. Report known overlaps and class counts; use no diagnostic labels for tuning. Benefits require downstream gains: lower PRM scores alone are insufficient, and small or inconsistent accuracy changes remain inconclusive. Reward hacking and policy distribution shift are residual risks.

Tentative timeline. Week 1: finalize the offline buffer, anchor audit, and PRM adaptation. Week 2: complete A–D for the first seed. Week 3: run the remaining paired seeds and reward diagnostics. Week 4: complete statistical analysis, error analysis, and the reproducible report.

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